

Unit 14 - Categorical Data

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The following B.E.S.T Standards will be covered in this unit:

MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies and determine possible associations in terms of a real-world context.

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

- *Clarification 1: Instruction includes discussions regarding the strengths and weaknesses of each data display.*
- *Clarification 2: Numerical univariate includes histograms, stem-and-leaf plots, box plots and line plots; numerical bivariate includes scatter plots and line graphs; categorical univariate includes bar charts, circle graphs, line plots, frequency tables and relative frequency tables; and categorical bivariate includes segmented bar charts, joint frequency tables and joint relative frequency tables.*
- *Clarification 3: Instruction includes the use of appropriate units and labels and, where appropriate, using technology to create data displays.*

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.

- *Clarification 1: Within the Probability and Statistics course, instruction includes the use of spreadsheets and technology.*

MA.912.DP.3.2 (Honors only) Given marginal and conditional relative frequencies, construct a two-way relative frequency table summarizing categorical bivariate data.

- *Clarification 1: Construction includes cases where not all frequencies are given but enough are provided to be able to construct a two-way relative frequency table.*
- *Clarification 2: Instruction includes the use of a tree diagram when calculating relative frequencies to construct tables.*
-

MA.912.DP.3.3 (Honors only) Given a two-way, relative frequency table or segmented bar graph summarizing categorical bivariate data, interpret joint, marginal and conditional relative frequencies in terms of a real-world context.

- *Clarification 1: Instruction includes problems involving false positives and false negatives.*

MA.912.DP.1.3 Explain the difference between correlation and causation in the contexts of both numerical and categorical data.

Unit 14, Lesson 1: Constructing Two-Way Tables



Benchmarks

MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies and determine possible associations in terms of a real-world context.

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.



Learning Targets

- ☐ I can construct a two-way frequency table.
- ☐ I can determine joint and marginal frequencies in a two-way frequency table.



14.1.1 Warm-Up: Percent Calculations

1. Newborn babies typically lose some weight in the first week after birth. Weight loss of up to 10% is common. If a newborn baby weighs 8.5 pounds at birth and loses 8% of their weight in the first week of life, how many pounds does the baby lose?
2. Anya earned a score of 75% on her last math quiz. If she earned 18 points, how many total points was the quiz worth?
3. Five hundred gallons of maple sap were boiled to become 12.5 gallons of maple syrup. What percentage of the 500 gallons of sap is the 12.5 gallons of maple syrup?



14.1.2 Guided Instruction: Two-Way Frequency Tables

A 2018 Pew Research Center survey of American teenagers was conducted to analyze trends in social media and technology.

741 teens who use social media were asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral).

1. The effect of social media is a
- ☐ quantitative variable.
 - ☐ categorical variable.

The data when asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral) is summarized.

- 219 said the effect was mostly positive
- 211 said the effect was mostly negative
- 311 said the effect was neutral

On the same survey, the Pew Research Center also collected data from teens on how much time they believe they spend with their friends in person (too much, too little, or about the right amount).

2. Time spent with friends in person (too much, too little, or just the right amount) is a
- ☐ quantitative variable.
 - ☐ categorical variable.

A **two-way frequency table** is a display used to summarize bivariate categorical data. The table can also be called a contingency table or a joint frequency table.

The two-way frequency table displays the results for teens when asked about the effect of social media, in addition to the results when being asked about time spent with friends in person.

Teen Social Media Survey

		Effect on People Your Age			
Time spent With friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

3. Circle the frequency for those who said the effect of social media was mostly positive.
4. Circle the frequency for those who said the effect of social media was mostly negative.
5. Circle the frequency for those who said the effect of social media was neutral.

6. How many teens reported spending too much time with friends in person? Circle the value in the table.
7. How many teens reported spending too little time with friends in person? Circle the value in the table.
8. How many teens reported spending just the right amount of time with friends in person? Circle the value in the table.

Each of the values circled is a **marginal frequency**.

The total frequency for one category of a categorical variable, which is the total frequency for any row or column in a two-way frequency table, is called a **marginal frequency**.

9. How many people reported that the effect of social media is mostly positive **and** reported that they spend too much time with friends in person?
10. How many people reported that the effect of social media is mostly negative **and** reported that they spend too much time with friends in person?
11. How many people reported that the effect of social media is mostly negative **and** reported that they spend too little time with friends in person?



In a two-way table, a **joint frequency** is the number of times a combination of two conditions occurs.

12. Draw a box around each joint frequency in the table.

13. What percentage of respondents reported that the effect of social media is mostly positive **and** reported that they spend too much time with friends in person?

$$\frac{\boxed{}}{\boxed{}} =$$

14. What percentage of respondents reported the effect of social media is mostly negative **and** reported that they spend too much time with friends in person?

$$\frac{\boxed{}}{\boxed{}} =$$



In a two-way table, a **joint relative frequency** is the ratio of the joint frequency and the total number of data points.



14.1.3 Collaboration: Interpreting a Two-Way Table

- With your partner, use the two-way frequency table to determine if each statement in the table is true or false. If a statement is false, correct the statement to make it true.

**Teen Social Media Survey
Effect on People Your Age**

Time spent With Friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

269 participants reported that they spend too little time with friends in person.	☐ True ☐ False
More teens reported that they spend too little time with friends than just the right amount with friends in person.	☐ True ☐ False
211 participants reported the effect of social media was mostly negative and spend just the right amount of time with friends in person.	☐ True ☐ False
186 participants reported that social media has a neutral effect and they spend just the right amount of time with their friends.	☐ True ☐ False
There was a total of 1,823 participants in the survey.	☐ True ☐ False



14.1.4 Guided Practice: Constructing a Two-Way Table

- Kieran decides to gather data on hair and eye color for his genetics project in science. Kieran's data is shown.

The table shows the number of people who match each description.

Brown hair, Brown Eyes	Brown Hair, Blue Eyes	Blonde Hair, Brown Eyes	Blonde Hair, Blue Eyes
33	18	11	24

- Construct a two-way frequency table to summarize the data.

			Total
Total			

- b. The total number of students surveyed with brown eyes is _____ .

This value represents a

- ☐ marginal frequency.
- ☐ joint frequency.

- c. The total number of students surveyed with brown eyes and blonde hair is _____ .

This value represents a

- ☐ marginal frequency.
- ☐ joint frequency.



14.1.5 Try It: Constructing a Two-Way Table

1. A survey of high school students asked participants if they play video games and what gender they identify as. The data is summarized:
- 68 participants identified as males who play video games
 - 60 participants identify as males who do not play video games
 - 145 participants do not play video games
 - There were 279 participants in the survey

- a. Complete the joint frequency table.

Student Video Game Activity

	Play Video Games	Do Not Play Video Games	Total
Male			
Female			
Total			

- b. How many students identify as female and do not play video games?
- c. Of the students who play video games, how many identify as female?
- d. How many students surveyed play video games?

2. Jackson County Schools, located in Marianna, Florida, conducted a student survey to find out how students spend the majority of their free time outside of school.

The data of this survey is shown in tables below.

Elementary Students		
On the Phone	Playing Video Games	Reading
389	863	1,242

Middle School Students		
On the Phone	Playing Video Games	Reading
532	557	481

High School Students		
On the Phone	Playing Video Games	Reading
978	844	224

Construct a two-way frequency table that shows all of the survey data.



14.1.6 Wrap-Up: Reflection

1. Rate your confidence in creating two-way tables to display bivariate categorical data.



0 1 2 3 4 5 6 7 8 9 10

2. What are some questions you still have about two-way frequency tables?

Unit 14, Lesson 2: Interpreting Joint and Marginal Frequencies



Benchmarks

MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies and determine possible associations in terms of a real-world context.

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.



Learning Targets

- ☐ I can determine joint and marginal frequencies.
- ☐ I can interpret joint and marginal frequencies in a real-world context.
- ☐ I can interpret a graphical display for bivariate categorical data.



14.2.1 Warm-Up: Bell Work

1. A survey conducted at Ambidextrous High School asked all 1,700 students to indicate their grade level and if they are left-handed or right-handed. Only 59 of the 491 freshmen are left-handed. Out of the 382 students in the sophomore class, 289 of them are right-handed. There are 433 students in the junior class and 120 of them are left-handed. There are 307 right-handed seniors.

Complete the two-way frequency table to display the results of the survey.

Dominant Hand Survey

					Total
Right-Handed					
Left-Handed					
Total					



14.2.2 Collaboration: Joint and Marginal Frequencies

1. In 2016, the Summer Olympics were held in Rio de Janeiro, Brazil. The United States of America won the most medals with a total of 121. China came in second, winning 70 total medals. This was a much higher total than the 9 medals China won two years later in the 2018 Winter Olympics held in Pyeongchang, South Korea. Norway led the medal count in 2018 with 39, a much higher number than the 4 total medals Norway won two years earlier in Brazil. Canada came in third in the 2018 Winter Olympics, winning 29 total medals.
 - a. Work with your partner to use the data regarding medal counts from the 2016 Summer Olympics and the 2018 Winter Olympics to make a two-way frequency table showing how the different countries' medal counts compare to each other.

	U.S.	China	Norway	Canada	Total
2016 Summer Olympics					217
2018 Winter Olympics					
Total	144				317

- b. How many total medals did the U.S. win in the 2016 and 2018 Olympics combined?

- c. Which team won the most medals in the two Olympics combined?
 - d. How many medals did Norway win in the 2018 Olympics?
 - e. Complete the statement.

_____ won the most medals in either the 2016 or 2018 Olympics, winning _____ medals in those games.

- f. What does the joint relative frequency, $\frac{121}{317}$, mean in this context?

2. Work with your partner to write questions that can be answered with the data in the table.
 - a. Write at least one question that must be answered with a joint frequency.
 - b. Write at least one question that must be answered with a marginal frequency.



14.2.3 Try It: Mono Lake

In California, there is a saltwater lake called Mono Lake. The salinity, or the concentration of salt in the lake, has varied over time. According to the Global Nature Fund, the salinity of Mono Lake was at an all-time high of 99 grams per liter (g/l) in 1982. The concentration of salt in Mono Lake increased as a result of water diversion. Streams that feed Mono Lake were diverted, and the fresh water was used for drinking water.

Regulations and compromises have helped control water diversion at Mono Lake since then, and the salinity of the lake is slowly returning to normal. In 2008, there were 79.8 grams of salt per liter in Mono Lake. As a result, the populations of two main organisms that live in the lake, brine shrimp and alkali flies, have increased to more optimal levels. In 1982, there were an estimated 2,000 brine shrimp per liter of lake water. In 2008, there were 3,000 brine shrimp per liter of lake water. In 1982, there were 1,200 alkali flies per liter of lake water. That number nearly tripled to 3,500 alkali flies per liter in 2008 when the salinity of Mono Lake was 79.8 g/l.

1. Construct a two-way frequency table that shows the alkali flies per liter of lake water and brine shrimp per liter of lake water living at Mono Lake when the salinity was 99 g/l and 79.8 g/l.

	Salinity of 99 g/l	Salinity of 79.8 g/l	Total
Brine Shrimp			
Alkali Flies			
Total			

2. Work with your partner to answer each question.

- How many alkali flies per liter of lake water were found in Mono Lake when the salinity was 99 g/l?
- When the salinity of Mono Lake was 79.8 g/l, how many brine shrimp and alkali flies per liter of lake water were found?
- Which is greater — the combined total of alkali flies, or the combined total of brine shrimp in Mono Lake from these two samples?
- What was the salinity of Mono Lake when the population of alkali flies was 3,500 per liter?
- What was the total amount of brine shrimp found in Mono Lake from the two studies?
- Was the population of brine shrimp smaller when the salinity of Mono Lake was 99 g/l or 79.8 g/l?



14.2.4 Your Turn!

- At Dupont Middle School in Jacksonville, FL, 635 students were asked which social media site they updated the most on a daily basis. Seventy-five seventh graders said they used Instagram the most. A total of 321 students said they updated Facebook the most. Of the 247 eighth graders, 23 of them updated Instagram the most. There are 181 sixth graders at Dupont. Of the 158 Twitter users, 45 of them were sixth graders. There were 123 eighth graders using Facebook.

Complete the two-way table summarizing the data.

Dupont Middle School Social Media Preference

	Facebook	Instagram	Twitter	Total
6 th				
7 th				
8 th				
Total				

2. Austin studies birth weight and height as part of his research at Florida School of Traditional Midwifery in Gainesville, Florida. The following two-way table shows data from births in the month of June 2021 at UF Health Shands Hospital.

Births from UF Health Shands Hospital (June 2021)

	Less Than 5 lbs.	5-8 lbs.	More Than 8 lbs.	Total
Less Than 16 in.	21		2	32
16-20 in.	14	97	68	
More Than 20 in.	0	8		
Total			101	

Use the table to answer the following questions and prompts.

- Circle the smallest marginal frequency in the table.
- Underline the largest joint frequency in the table.
- Describe what the value 14 means in context of the data.
- Describe what the value 101 means in context of the data.



14.2.5 Wrap-Up: Ticket Out the Door

Use the two-way table showing students' field trip preferences to answer the questions.

Students' Field Trip Preferences

	Amusement Park	Water Park	Mystery Destination	Total
Freshmen	25	9	20	54
Sophomores	35	6	5	46
Total	60	15	25	100

- Underline the smallest joint frequency.
- Explain what the smallest joint frequency represents in terms of students' field trip preferences.
- Circle the largest marginal frequency.
- Explain what the largest marginal frequency represents in terms of students' field trip preferences.

Unit 14, Lesson 3: Two-Way Tables and Segmented Bar Graphs



Benchmarks

MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies and determine possible associations in terms of a real-world context.

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.



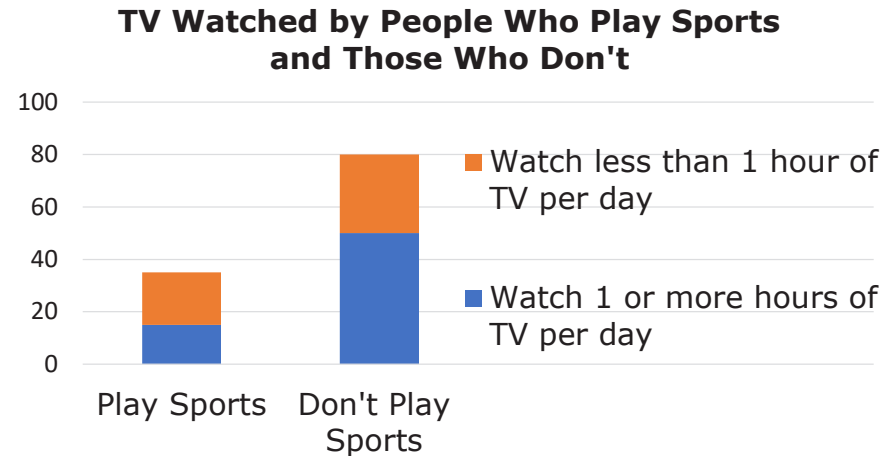
Learning Targets

- ☐ I can correctly match a segmented bar graph and data in a two-way table.
- ☐ I can interpret a graphical display for bivariate categorical data.



14.3.1 Warm-Up: Notice and Wonder

A segmented bar graph is shown.



1. What do you notice about the display?
2. What do you wonder?



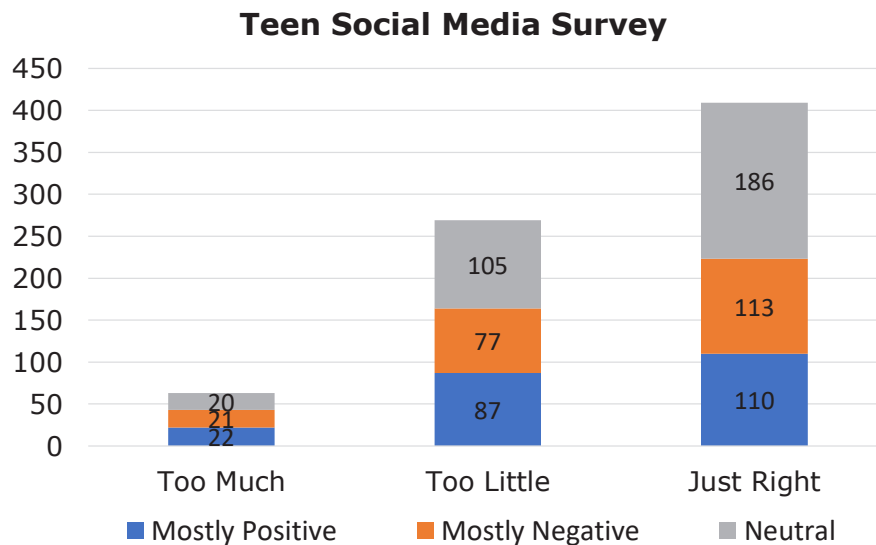
14.3.2 Exploration: Segmented Bar Graphs

Bivariate categorical data can be displayed in a two-way frequency table as shown:

Teen Social Media Survey

		Effect on People Your Age			
Time spent With Friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

It can also be displayed in a **segmented bar graph**.



Three two-way frequency tables and four segmented bar graphs are shown.

Table 1

Peoples' Social Media Preference by Age Group

	Less Than 18 Years Old	18 – 30 Years Old	Over 30 Years Old	Total
Snapchat	147	72	90	309
Facebook	23	56	112	191
Total	170	128	202	500

Table 2

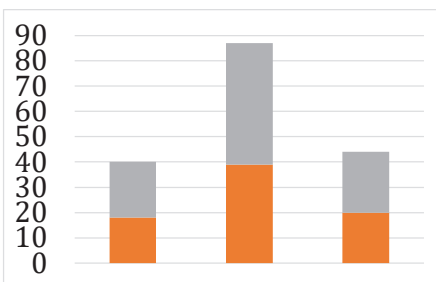
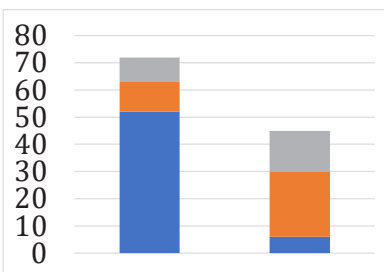
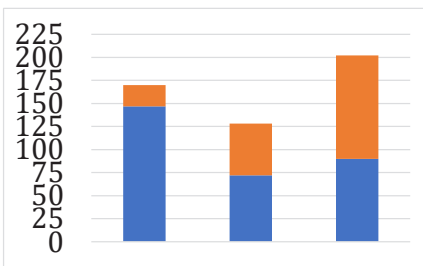
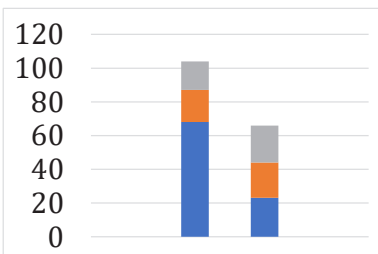
Points Scored During a Volleyball Match

	University of Miami, Florida	Jacksonville University, Florida	Total
Spike Kills	18	22	40
Service Aces	39	48	87
Other	20	24	44
Total	77	94	171

Table 3

Emotional States Related to Meditation Practice

	Meditated	Did Not Meditate	Total
Calm	52	6	58
Excited	11	24	35
Agitated	9	15	24
Total	72	45	117

Graph A**Graph B****Graph C****Graph D**

1. Work with your partner to match each two-way frequency table to the segmented bar graph that displays the same data.

Two-Way Frequency Table	Segmented Bar Graph
1	☐ A ☐ B ☐ C ☐ D
2	☐ A ☐ B ☐ C ☐ D
3	☐ A ☐ B ☐ C ☐ D

2. Work with your partner to label the bars and their sections in the segmented bar graphs so that the categories represented by each bar are indicated.
3. Discuss with your partner how joint frequencies are displayed in a segmented bar graph. Summarize your discussion.



14.3.3 Bring It Together: Representing Bivariate Categorical Data With Segmented Bar Graphs

Another way to represent the bivariate categorical data in a two-way frequency table is with a segmented bar graph.

1. In a two-way frequency table, marginal frequencies are found in the

☐ center
☐ margins

 of the table. These numbers

express the

☐ totals
☐ parts

 of each category.

2. Carolyn concluded that marginal frequencies are represented by the height of each bar on a segmented bar graph.

Bailey argued that the marginal frequencies are represented by the sum of the segment heights with the same color for each color.

Which student do you agree with? Explain your reasoning.

3. In a two-way frequency table, joint frequencies are found in the

☐ center
☐ margins

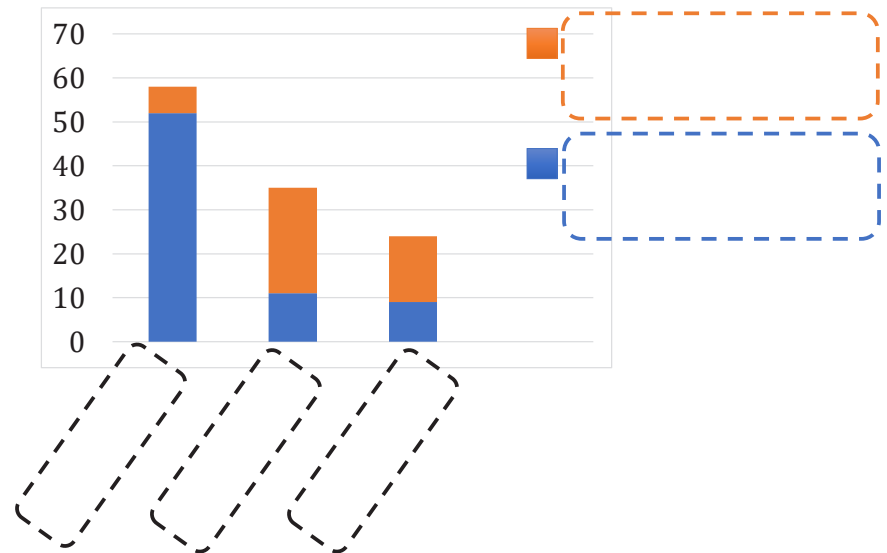
 of the table. Joint frequencies are represented by the

- ☐ total height of a bar.
☐ segments of different colors within each bar.

4. Consider the data in the table and segmented bar graph.

People's Emotional States Related to Meditation Practice

	Meditated	Do not Meditate	Total
Calm	52	6	58
Excited	11	24	35
Agitated	9	15	24
Total	72	45	117



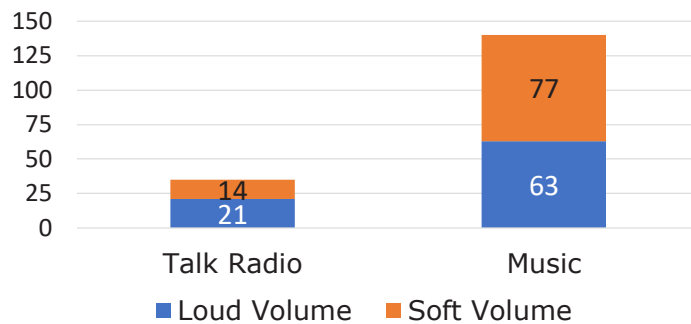
- a. Circle the marginal frequencies in the table.
b. Label each marginal frequency in the dotted boxes.
c. Label the value for each joint frequency.



14.3.4 Try It: Interpreting Segmented Bar Graphs

- Long-haul truckers were recently surveyed at a rest area in Tallahassee, Florida. The truckers were asked about what and how they primarily listen to audio in their trucks while on the road.

Long-Haul Truck Driver Listening Preferences



Determine whether each of the following statements is true or false. If the statement is false, correct the statement to make it true.

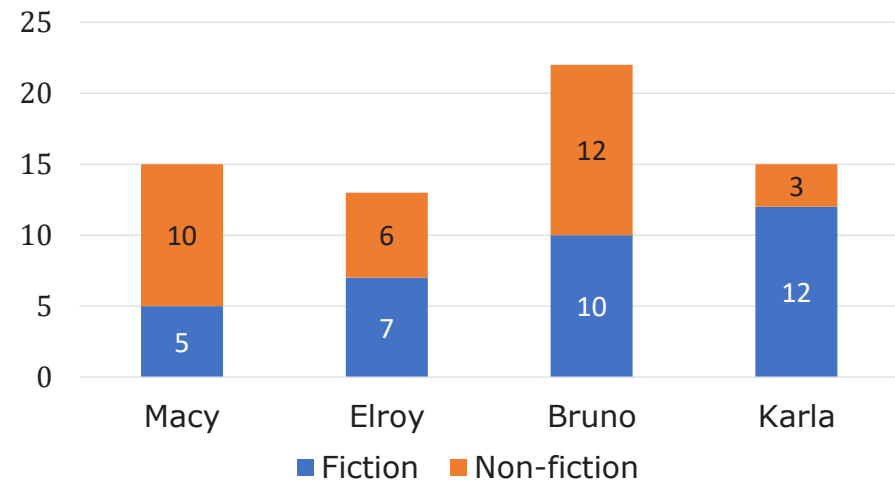
14 of those surveyed prefer loud volume.	☐ True ☐ False
140 of those surveyed prefer soft volume.	☐ True ☐ False
63 of those surveyed prefer music at a soft volume.	☐ True ☐ False
More people prefer music than talk radio.	☐ True ☐ False
More people prefer talk radio to be at a loud volume than at a soft volume.	☐ True ☐ False



14.3.5 Wrap-Up: Ticket Out the Door

Macy, Elroy, Bruno, and Karla each kept track of the books they read for a year. Their data is represented in the segmented bar graph shown. Use the segmented bar graph to answer the questions.

Books Read



- What does the value 7 mean in this context?
- How many fiction books were read?

Unit 14, Lesson 4: Determining Possible Associations in Data



Benchmark

MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate categorical data. Interpret joint and marginal frequencies and determine possible associations in terms of a real-world context.



Learning Target

- ☐ I can determine possible associations from a two-way table and segmented bar graph.



14.4.1 Warm-Up: Playing Instruments and Sports

Orlando asked his classmates these questions:

- Do you play a sport?
- Do you play a musical instrument?

1. The two-way table gives partial results from the survey. Complete the table, assuming that all students answered both questions.

	Plays Instrument	Does not Play Instrument	Total
Plays Sport	5		16
Does not Play Sport			
Total		15	25

2. What fraction of the total students do not play an instrument and also do not play sports?
3. Does the fraction you found in represent a joint or marginal relative frequency?



14.4.2 Guided Instruction: Finding Possible Associations in the Data

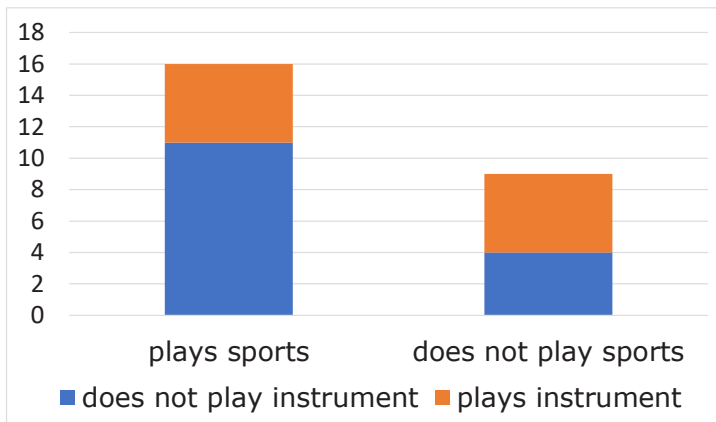
Two-way tables are effective for organizing data. Creating a segmented bar graph for the data in a two-way table can help make possible associations more visible.



An **association** is a way to describe the relationship between the two variables in a bivariate data set. For categorical data, descriptions include strong or weak.

Orlando would like to know if there is any association between playing an instrument and playing a sport. He makes a segmented bar graph from the data in the two-way table.

	Plays Instrument	Does not Play Instrument	Total
Plays Sports	5	11	16
Does not Play Sports	5	4	9
Total	10	15	25



1. Orlando and Hannah discussed possible associations in the data using the displays. Each student's thinking is shared in the table.

Orlando's Thinking	Hannah's Thinking
"Of the students who play sports, there are 5 people who play an instrument. This is also true of the students who do not play sports. This means there is no association, or relationship, between playing sports and playing an instrument because it's 5 people in both cases."	"While there are only 5 people in each category who play an instrument, there is a much larger portion of the students who do not play sports. Half of the students who don't play sports do play an instrument. I think there is an association between playing sports and playing an instrument. It seems more likely a student plays an instrument if they don't play a sport."

- a. Without the help of your partner, explain whether you agree more with Orlando or Hannah.
- b. Compare your answer with your partner's. Make any edits to your explanation based on your discussion.



14.4.3 Bring It Together: Using Segmented Bar Graphs to Find Possible Associations

1. Complete the statements.
 - a. Segmented bar graphs are useful for displaying categorical ☐ univariate
☐ bivariate data.
 - b. Segmented bar graphs can be helpful for making visual comparisons between joint and marginal frequencies. Take, for example, the bar on the left of the graph from the previous component labeled “plays sports.”

About ☐ $\frac{1}{2}$
☐ $\frac{1}{3}$ of this bar is orange. The other bar, labeled “does not play sports” is just over ☐ $\frac{1}{2}$
☐ $\frac{1}{3}$ orange. This means a larger portion of the students who ☐ play a sport
☐ do not play a sport play an instrument, which means there may be an association between playing sports and instruments.

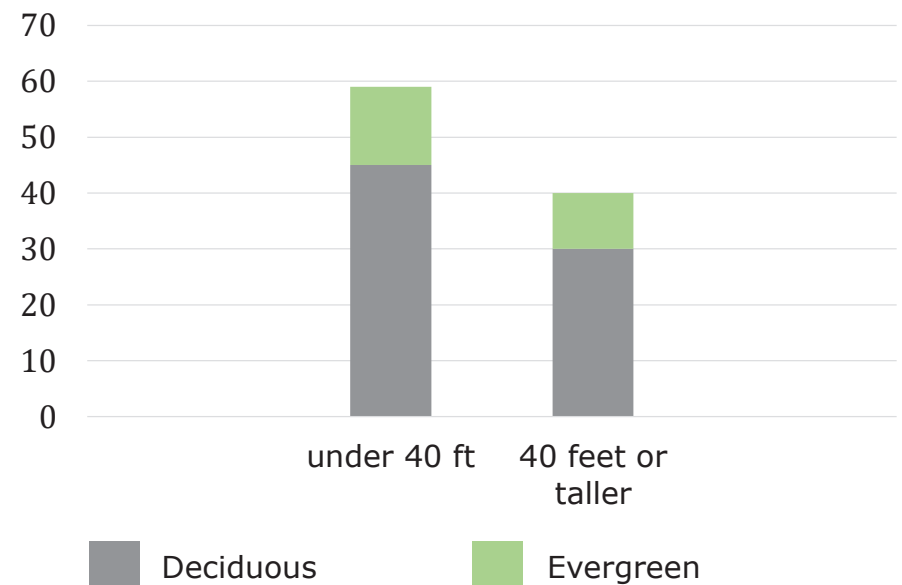


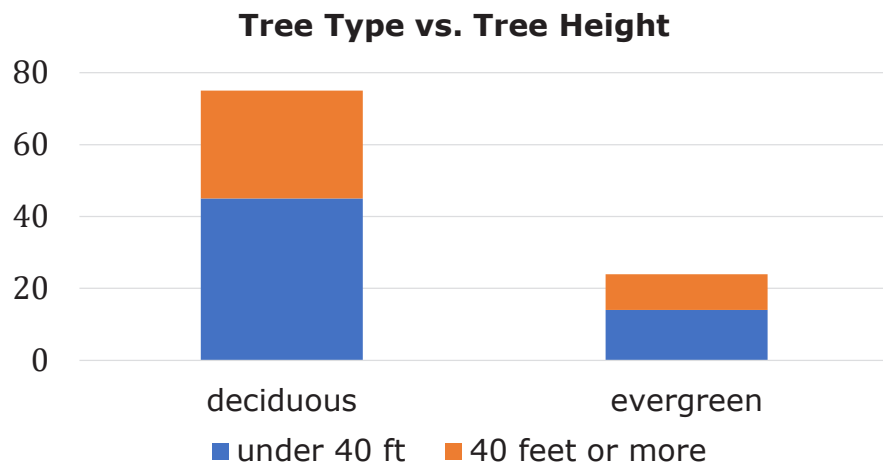
14.4.4 Try It: Possible Associations

1. An ecologist is studying a forest with a mixture of tree types. Since the average tree height in the area is 40 feet, the height of each tree is measured against that and recorded. The type of tree is also recorded. The results are shown in the table and segmented bar graphs.

	Under 40 Feet	40 Feet or Taller	Total
Deciduous	45	30	75
Evergreen	14	10	24
Total	59	40	99

Tree Height vs. Tree Type



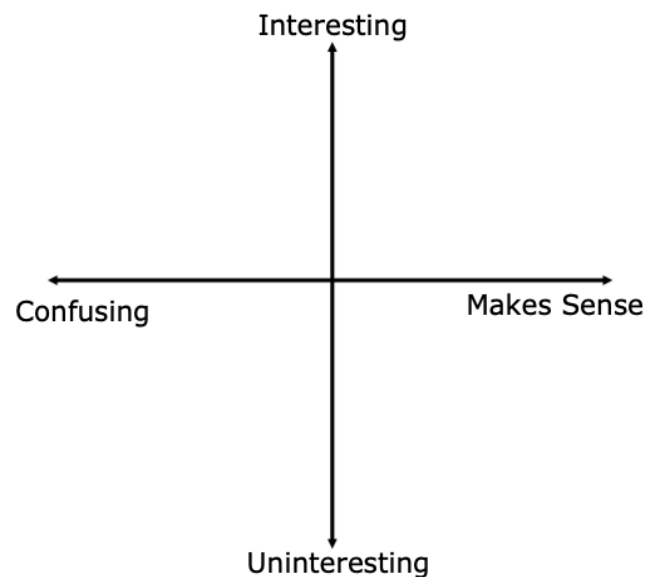


- Explain whether or not there is evidence of an association between tree height and tree type.
- Explain which data display was more helpful in determining whether or not there is evidence of an association between height and type of tree.



14.4.5 Wrap-Up: Reflection

- Plot a point in one of the quadrants below to indicate where you are on today's learning target, "I can determine possible associations from a two-way table and segmented bar graph."



Unit 14, Lesson H1: Creating Relative Frequency Tables



Benchmarks

MA.912.DP.3.2 Given marginal and conditional relative frequencies, construct a two-way relative frequency table summarizing categorical bivariate data.

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical and whether it is univariate or bivariate, and interpret the different components and quantities in the display.



Learning Target

☐ I can construct a two-way relative frequency table.



14.H1.1 Warm-Up: What Percentage?

1. Mindy is in charge of ordering supplies for the break room at her workplace, so she surveys her coworkers. She asks them if they prefer coffee or tea and whether or not they like sugar in their beverages. Her findings are shown in the two-way table.

	Coffee	Tea	Total
Sugar	15	8	23
No Sugar	25	2	27
Total	40	10	50

Use the two-way table to answer the following questions.

- a. What percentage of people surveyed prefer coffee?
- b. What percentage of people surveyed drink tea with sugar?



14.H1.2 Exploration: Two-Way Relative Frequency Tables

A 2018 Pew Research Center survey of American teenagers was conducted to analyze trends in social media and technology. 741 teens who use social media were asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral) and how much time they believe they spend with their friends in person (too much, too little, or about the right amount). The results are summarized in the joint frequency table.

Time spent With Friends in Person	Effect on People Your Age				
		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

The data can also be displayed in a joint relative frequency table, where the cells represent the joint and marginal relative frequencies. These are often useful to construct when given percentages for a data set, rather than frequencies.

Time spent With Friends in Person	Effect on People Your Age				
		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	2.97%	2.83%	2.7%	8.5%
	Too Little	11.74%	10.39%	14.17%	36.3%
	Just Right	14.84%	15.25%	25.10%	55.20%
	Total	29.55%	28.48%	41.97%	100.00%



A **joint relative frequency** is the ratio of a joint frequency and the total number of data points.

A **marginal relative frequency** is the ratio of a marginal frequency and the total number of data points.

1. What is the sum of the joint relative frequencies?
2. What is the sum of the marginal relative frequencies for the variable "effect on people your age?"
3. What is the sum of the joint relative frequencies for those who believe the effect of social media is mostly positive?

4. What is the sum of the joint relative frequencies for those who believe they spend too much time with friends in person?
5. Use only the values from the joint frequency table to answer the following questions.
- What percentage of those who spend too much time with friends in person believe that the effect of social media has been mostly positive?
 - What percentage of those who spend too much time with friends in person believe that the effect of social media has been mostly negative?

The percentages calculated in Parts A and B above are **conditional relative frequencies**.



A **conditional relative frequency** is the ratio of a joint relative frequency to a marginal relative frequency. Equivalently, the ratio of a joint frequency and a marginal frequency.



14.H1.3 Guided Instruction: Creating a Two-Way Relative Frequency Table

Consider the following data on whether or not animals received a microchip from the Kirkman Road Veterinary Clinic's database of canines (dogs) and felines (cats).

- 37.61% of animals in the sample were felines.
- 30.63% of animals in the sample did not have microchips.
- Of the animals with microchips, 81.17% were canines.
- 34.73% of the felines had microchips.

1. Complete each statement.

a. 37.61% represents a frequency.

- ☐ joint
- ☐ marginal
- ☐ conditional

relative

b. 30.63% represents a frequency.

- ☐ joint
- ☐ marginal
- ☐ conditional

relative

c. 81.17% represents a frequency.

- ☐ joint
- ☐ marginal
- ☐ conditional

relative

d. 34.73% represents a frequency.

- ☐ joint
- ☐ marginal
- ☐ conditional

relative

2. Explain how to determine the relative frequency of microchipped canines in the sample.
3. The value in question 2 represents a relative frequency.

☐ joint
☐ marginal
☐ conditional
4. Use the data to create a joint relative frequency table. Space is provided to make any calculations.

Clients at Kirkman Road Veterinary Clinic

	Microchipped	Not Microchipped	Total
Feline			
Canine			
Total			

5. Complete the statement.

The smallest percentage of animals in the sample were

- ☐ microchipped
☐ not microchipped

- ☐ felines.
☐ canines.

6. What percentage of canines had microchips?

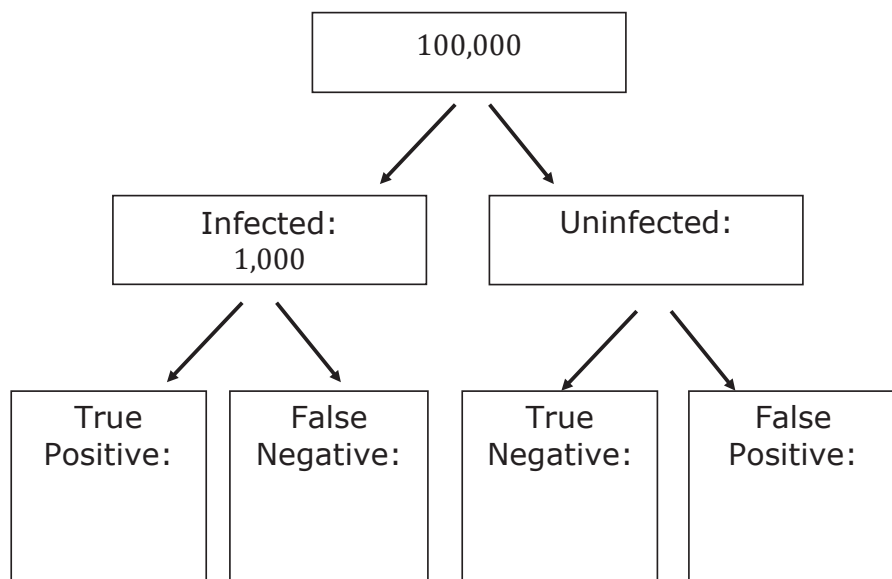
7. Which kind of pet is more likely to be microchipped? Explain your reasoning.



14.H1.4 Guided Practice: Creating a Two-Way Relative Frequency Table

A study is performed to determine the efficacy of a new test that determines whether or not a patient has a disease. 100,000 people were included in the study. 1% of the participants are infected with the disease. Of those infected, the test correctly identified the presence of the disease in 750 patients, and 250 patients received a false negative. For those patients that were not infected with the disease, the test was correct in 98,505 cases.

1. Use this data to complete the tree diagram.



2. Complete the two-way relative frequency table to represent the data.

	Positive	Negative	Total
Infected			1%
Not Infected			99%
Total			



14.H1.5 Try It: RMS Titanic

- The RMS Titanic was a British passenger liner operated by the White Star Line that sank in the North Atlantic Ocean on April 15, 1912, after striking an iceberg during her maiden voyage from Southampton to New York City. Of the estimated 2,201 passengers and crew aboard, 1,490 died, making the sinking, at the time, one of the deadliest of a single ship.

Passengers can be classified as whether the person survived (Yes or No) and the class of the ticket they had (1st, 2nd, 3rd, or Crew).

- 14.8% of passengers had 1st class tickets.
- 13.0% of passengers had 2nd class tickets.
- 35.4% of those who died were third class passengers.
- 25.0% of those who survived were 3rd class passengers.
- 76.0% of the crew died.
- 62.5% of 1st class passengers survived.

Create a two-way relative frequency table based on the data provided.

Ticket	Survived		Total
	No	Yes	
1st			
2nd			
3rd			
Crew			
Total			



14.H1.6 Wrap-Up: Reflection

- Complete each statement.
 - After today's lesson, I am still unsure about ...
 - To fill this gap, I intend to ...

Unit 14, Lesson H2: Interpreting Relative Frequency Tables and Segmented Bar Graphs



Benchmarks

MA.912.DP.3.2 Given marginal and conditional relative frequencies, construct a two-way relative frequency table summarizing categorical bivariate data.

MA.912.DP.3.3 Given a two-way relative frequency table or segmented bar graph summarizing categorical bivariate data, interpret joint, marginal, and conditional relative frequencies in terms of a real-world context.

MA.912.DP.1.3 Explain the difference between correlation and causation in the contexts of both numerical and categorical data.



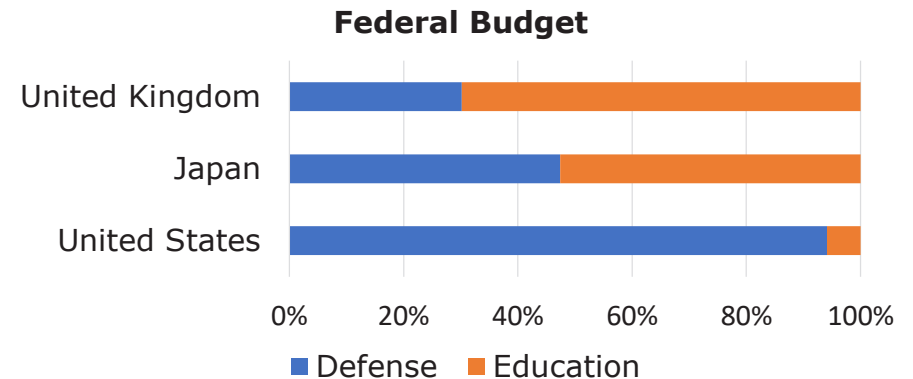
Learning Targets

- ☐ I can construct a two-way relative frequency table.
- ☐ I can interpret joint, marginal, and conditional relative frequencies in a real-world context.



14.H2.1 Warm-Up: Notice and Wonder

The segmented bar graph represents the distribution of where some of the 2009 federal budgets were expected to go.



1. What do you notice?

2. What do you wonder?



14.H2.2 Collaboration: Two-Way Relative Frequency Tables

Data was collected from a 2019 random sample of traffic accidents in Chicago. Data on the road conditions at the time of the accidents (dry, wet/slushy, icy) and the amount of damage caused (\$500 or less, \$501–\$1,500, more than \$1,500) is summarized.

- 27.76% of the accidents involved damage of \$500 or less.
- 28.77% of the accidents involved damage of \$501 to \$1,500.
- 78.43% of the accidents involved dry road conditions.
- 20.40% of the accidents involved wet/slushy road conditions.
- Of the accidents with damage of more than \$1,500, 69.77% involved dry road conditions.
- Of the accidents with damage of \$501–\$1,500, 80.22% involved dry road conditions.
- 24.79% of the accidents with icy road conditions had damage of \$501–\$1,500.
- 10.26% of the accidents with icy road conditions had damage of \$500 or less.

1. Use the data provided to construct a two-way relative frequency table. Round values to the nearest hundredth.

Traffic Accident Data

Road Conditions	Damage			Total
	\$500 or less	\$501–\$1,500	More than \$1,500	
Dry				
Wet/Slushy				
Icy				
Total				

2. Interpret the joint relative frequency for dry road conditions with damage of \$500 or less.
3. Interpret the marginal relative frequency for traffic accidents that had icy road conditions.

4. When the road conditions are icy, is the likelihood of the different amounts of damage the same? Explain your reasoning using conditional relative frequencies.

5. Does there appear to be an association between road conditions and the amount of damage caused?

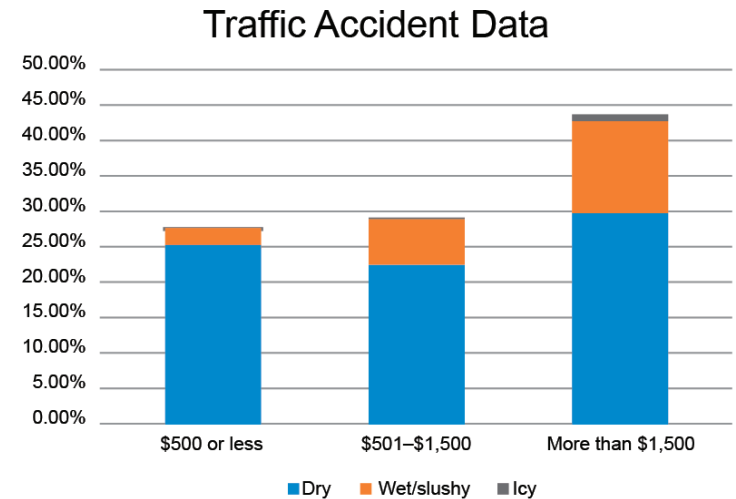
6. Does this mean that more hazardous road conditions cause more damage in traffic accidents?



Association does not imply causation!
If there is an association between two variables, it does not mean that one causes the other.
Two variables can be associated without having a causal relationship.

Another way to represent bivariate categorical data from a two-way relative frequency table is with a segmented bar graph. These graphs are helpful in visualizing comparisons between different variables by identifying how much larger or smaller different parts of each bar are compared to other bars.

The data from the two-way frequency table can be displayed in a segmented bar graph as shown.



7. With your partner, discuss the similarities and differences between the two-way relative frequency table and the segmented bar graph. Summarize your discussion.



14.H2.3 Try It: Interpreting Relative Frequencies

The General Social Survey (GSS) interviews a random sample of Americans every two years. In one question, respondents are asked to rate their general happiness level. Sample results for 1972, 1998, and 2021 are summarized in the table.

	1972	1998	2021	Total
Very Happy	486	891	783	2,160
Pretty Happy	855	1,575	2,308	4,738
Not Too Happy	265	340	923	1,528
Total	1,606	2,806	4,014	8,426

1. After reviewing the two-way frequency table, Diana said, "It seems like people in 2021 were a lot happier than in 1972."
 - a. Why do you think Diana said this?
 - b. Explain why this may not be an accurate conclusion to draw from the data.

2. Use the data to create a two-way relative frequency table. Round values to the nearest tenth of a percent.

	1972	1998	2021	Total
Very Happy				
Pretty Happy				
Not Too Happy				
Total				

3. Interpret the value 9.3% in context of the data.
4. Interpret the value 33.3% in context of the data.
5. Interpret the value 25.6% in context of the data.

6. Interpret the value $\frac{27.4\%}{47.6\%}$ in context of the data.

7. What percentage of the respondents from 1972 reported they were very happy?

8. What percentage of the respondents from 1998 reported they were very happy?

9. What percentage of the respondents from 2021 reported they were very happy?

10. What can be concluded about the general happiness of Americans from the two-way relative frequency table?



14.H2.4 Wrap-Up: Ticket Out the Door

1. The RMS Titanic was a British passenger liner operated by the White Star Line that sank in the North Atlantic Ocean on April 15, 1912, after striking an iceberg during her maiden voyage from Southampton to New York City.

The two-way relative frequency table classifies the passengers as whether the person survived (Yes or No) and the class of the ticket they had (1st, 2nd, 3rd, or Crew).

Ticket	Survived		Total
	No	Yes	
1st	5.5%	9.2%	14.8%
2nd	7.6%	5.4%	12.9%
3rd	24.0%	8.1%	32.1%
Crew	30.6%	9.6%	40.2%
Total	67.7%	32.3%	100.0%

a. What does the value 30.6% mean in context of the data?

b. What does the value $\frac{30.6\%}{40.2\%}$ mean in context of the data?